



Term Project Stage 2

Team 04

ADA University

Spring 2026 Business Process Modeling for IT Solutions (INFT-3610 - 20301)

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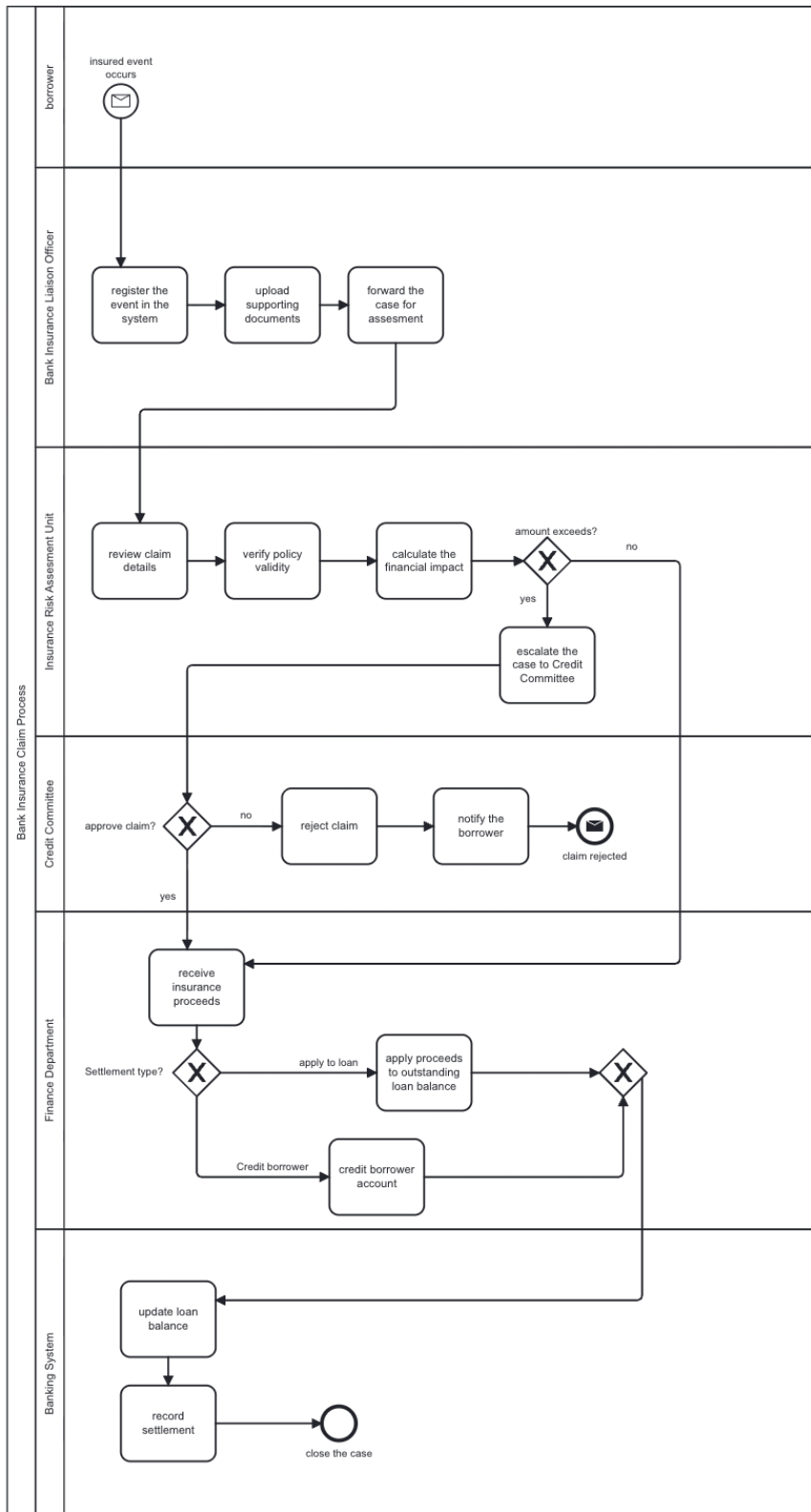
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INTRODUCTION

This report provides an analysis of the Insurance Event Application Process for the internal handling of insured events relating to mortgaged properties that are required to insure their property as part of their loan agreement. This Stage 2 Report builds upon the AS-IS model developed in Stage 1, further enhancing our understanding of the process model, verifying its correctness using formal verification methods, mapping organizational responsibility to each activity, and defining all data elements used during the process. The objective is to develop an organized, formalized understanding of the overall process as it relates to both BPMN modelling principles and organizational reality of the bank.

TASK 1. AS-IS PROCESS MODEL & STAGE 1 REFLECTION

TASK 1.1. STAGE 1 MODEL



TASK 1.2. STAGE 1 REFLECTION

The model provided above shows the current state of the process with no alterations made to the drawing. The model has one "Bank Insurance Claim Process" pool representing the name of the process that is being performed. This is a reasonable name for the pool, however, based on BPMN best practices, a pool typically indicates an entity who is part of a process. An example of an alternative and more common way to specify the name of the pool would be to give it the name of the organization managing the process in this case, "Bank". This clarifies that the pool is the conducting entity of the activities being executed, as opposed to the actual process itself.

Every internal process participant, known as bank organizational units, is displayed in the pool as a swimlane. A swimlane is an uninterrupted row that represents a particular role or an organizational unit that is responsible for the execution of defined or designated activities. Roles or organizational units represented include "Borrower", "Bank Insurance Liaison Officer", "Insurance Risk Assessment Unit", "Credit Committee", "Finance Department" and "Banking System". Each of the activities is assigned to only one swimlane, so the responsibility for performing each of the prescribed activities is clearly defined without duplicate responsibilities.

The "Insurance Partner" has not been displayed on the model, which is acceptable because it is also an external entity and it is not necessary to know the detail of how to interact with the "Insurance Partner" in order to understand the internal bank processes. The "Borrower" and the "Insurance Partner" may also be represented as separate pools to show the inter-organizational communication between them and the bank using message flows.

In the current process model, we have four XOR gateways. The first XOR split gateway displays the alternative pathways that could be taken after the "calculate financial impact" activity, since the amount either exceeds the limits and the case escalates to the

Credit Committee for approval or the amount does not exceed the predefined limits and the Finance Department proceeds. While reviewing the current model, we identified that additional XOR join gateway should have been used before the “receive insurance proceeds” activity, because once we choose any of the two possible pathways, we have to merge them with an XOR join gateway if they do not lead to separate end events, and lead to one activity instead.

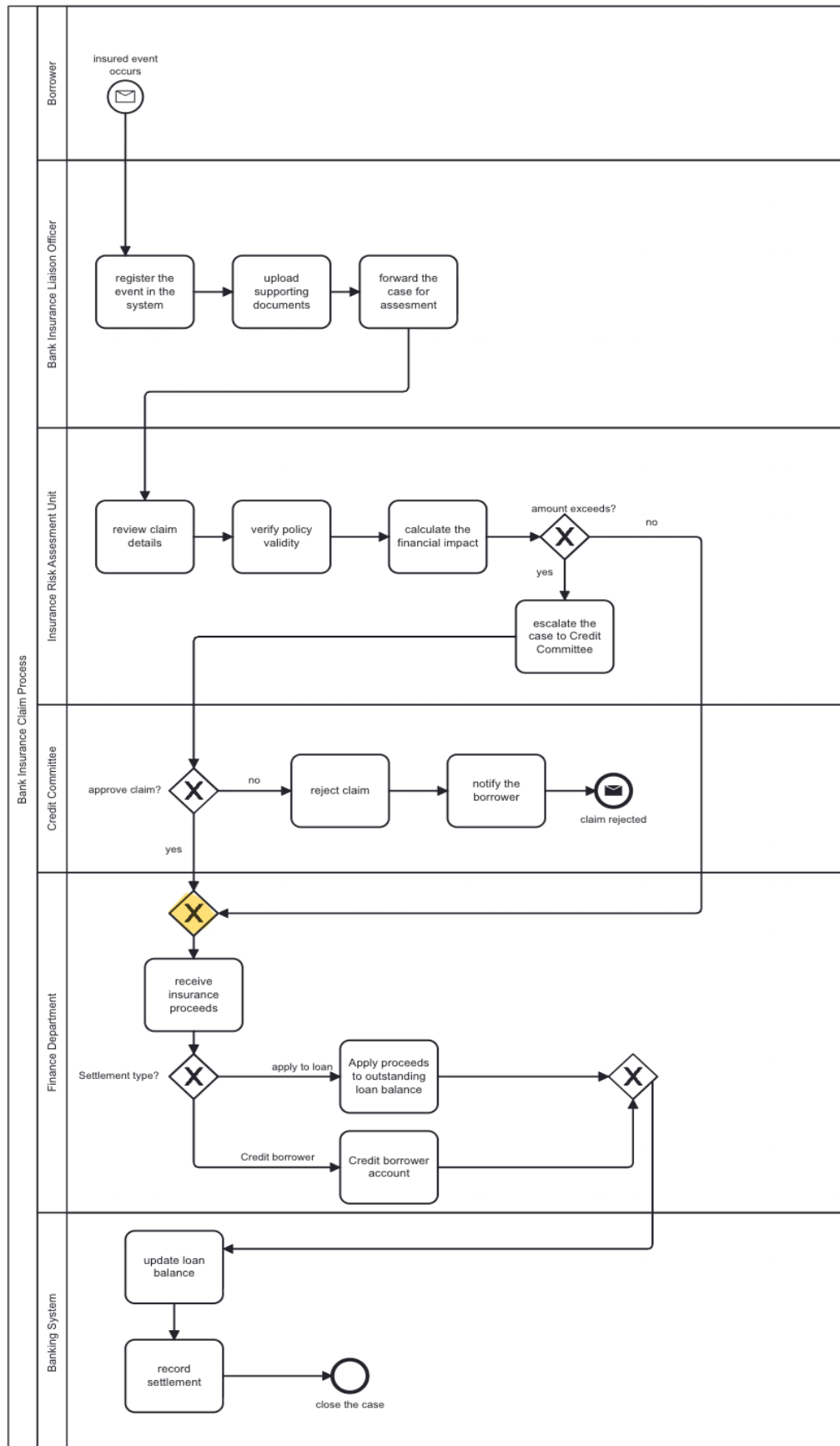
The second XOR split gateway appears in the “Credit Committee” lane that is responsible for either approving the claim or rejecting it. We do not need to merge them because the decision leads to a separate end event, if the claim is rejected, but if the claim is approved the “Finance Department” can proceed.

We notice the third XOR gateway appearing in the “Finance Department” lane. This gateway represents the logic of either reducing the loan balance if the loan balance is outstanding or crediting the borrower account. We can see that this split XOR gateway was merged with a join XOR gateway, since there are no separate end events and the process must continue with “update loan balance” activity. There is no other gateway type used in this process model, since we do not have any parallel executing activities in the process case description.

In the Stage1 process model, we have two end events which is consistent with the process logic. The first end event occurs when the Credit Committee decides to reject the claim and proceeds with notifying the Borrower about this decision and the end event “claim is rejected” closes this case. Otherwise, the case continues with Finance Department receiving the claim and assessing it further. Once the process finishes with updating the loan balance and recording the settlement, the end event "close the case" is used to denote the end of the case.

After completing the analysis of Stage1 model, we decided to modify it slightly by adding the highlighted in the diagram below XOR join gateway to the diagram before the “receive insurance proceeds” activity to comply with the logic of the process. No other modifications were made to the overall configuration of the diagram because there are no errors within the overall activities sequentially and logically as documented in the original model. All of the activities are further clearly identified with their respective organizational units, and all of the gateways and end events of the model continue to appropriately depict the decision-making logic and expected results for the process.

TASK 1.3. AS-IS PROCESS MODEL



TASK 2. RESOURCE ALLOCATION ANNOTATION

The roles and responsibilities determined in Stage 1 are the basis of the resource allocation of the AS-IS process. Every role identified during the initial analysis has a set of activities assigned to it and these activities are now explicitly translated into the process model. In order to have a clear and structured picture of the way the responsibilities are distributed, a resource allocation table and RACI chart are utilized. This strategy has the advantage of making each activity well attached to an organizational role or system, which shows the reality in the way the process is actually being carried out.

TASK 2.1. IDENTIFICATION OF HUMAN/AUTOMATED ACTIVITIES

Three types of execution are noted among the activities in terms of type of resources used:

Human roles only do some activities like checking details of claims, checking the validity of the policy, determining the financial effects and accepting or rejecting the claim. There are others that can be done only by the Banking System as an automated resource like the updating of the loan balance, and the recording of the settlement that do not necessitate human intervention and are automatically executed when the corresponding trigger is received. A third category is a combination of both that a human role makes decision or authorization, and the Banking System performs it. The notification of the borrower, application of proceeds to the balance in the loan, and credit of the borrower account follow this pattern.

TASK 2.2. COLLABORATIVE TASKS

There were no defined activities that would involve simultaneous involvement of two or more roles in a manner that would demand a specific collaborative swimlane. Even though some of these activities require a human element and the Banking System like informing the

borrower or adding the proceeds to the loan balance, these are not collaborative tasks in the BPMN sense. In both of them, the human role starts or approves the action, and the Banking System performs it in a sequence. It is a handoff but not concomitant joint participation, hence no special collaborative lane is needed. Moreover, although the approval decision of the Credit Committee is made by a group of people, who discuss the issue internally, the committee is rightly represented as one organizational unit in one swimlane, which is a collective role within a single decision-making authority and not a group of participants acting independently. There is consequently no duplication of activities across the lanes.

TASK 2.3. RESOURCE ALLOCATION PATTERNS

Everything in this process is allocated on a role basis. Every action is allocated to the specific organizational duties like the Bank Insurance Liaison Officer, the Insurance and Risk Assessment Unit, the Credit Committee, the Finance Department, and the Banking System as opposed to the specific named individuals. In this model, any qualified member of the role can be available to perform the activity, and a member can perform and take it. This is applicable both to human functions and to the Banking System as an automated resource of the organization.

Direct allocation where a task had to be given to a named person as there was unique personal knowledge or expertise was not found in this process. Every activitie can be done by any qualified individual or system holding the similar role in the organization.

TASK 2.4. RESOURCE ALLOCATION TABLE

Activity	Responsible Role/Resource	Resource Type	Allocation Type	Justification
Register the event in the system	Bank Insurance Liaison Officer	Human	Role-based	This role is responsible for initiating the claim process.
Upload supporting documents	Bank Insurance Liaison Officer	Human	Role-based	Document collection and upload duties of roles'
Forward the case for assessment	Bank Insurance Liaison Officer	Human	Role-based	The officer ensures the case reaches the assessment unit
Review claim details	Insurance and Risk Assessment Unit	Human	Role-based	This unit is responsible for reviewing claims
Verify policy validity	Insurance and Risk Assessment Unit	Human	Role-based	Policy verification is a standard responsibility
Calculate financial impact	Insurance and Risk Assessment Unit	Human	Role-based	Financial analysis is performed by this unit as a part of assessment
Escalate the case to Credit Committee	Insurance and Risk Assessment Unit	Human	Role-based	Escalation is triggered based on internal rules
Approve or reject claim	Credit Committee	Human	Role-based	The committee has decision-making authority
Notify the borrower	Credit Committee + Banking System	Human+ Automated System	Role-based	The committee initiates the rejection, the Banking System generates and sends the notification
Receive insurance proceeds	Finance Department	Human	Role-based	Finance handles incoming funds
Apply proceeds to outstanding loan balance	Finance Department + Banking System	Human + Automated System	Role-based	Finance decides the application, system executes adjustment
Credit borrower account	Finance Department + Banking System	Human + Automated System	Role-based	Finance authorizes the credit, the system performs the transaction
Update loan balance	Banking System	Automated System	Role-based	Fully automated system task triggered upon completion of settlement action
Record settlement	Banking System	Automated System	Role-based	System records and finalizes the process

TASK 2.5 RACI CHART

RACI chart is implemented to explain the distribution of responsibilities among the participants of the process. It determines who will do the work, who is to make the decision and who will know the result. The four names are:

- **R (Responsible)** – performs the task
- **A (Accountable)** – has final authority and decision power
- **C (Consulted)** – provides input or expertise when needed
- **I (Informed)** – is notified about the outcome

Even though the RACI model has the role of Consulted (C), no actions in the process involved consultative steps that needed to be taken explicitly. Although some of these roles like the Borrower or the Risk Assessment Unit might be viewed as input in some circumstances, they are already inherent in the specified activities instead of being modelled as distinct consultation procedures.

The BPMN model does not specifically depict consultation as a specific step hence C role is not a part of the RACI chart which is consistent with the process model.

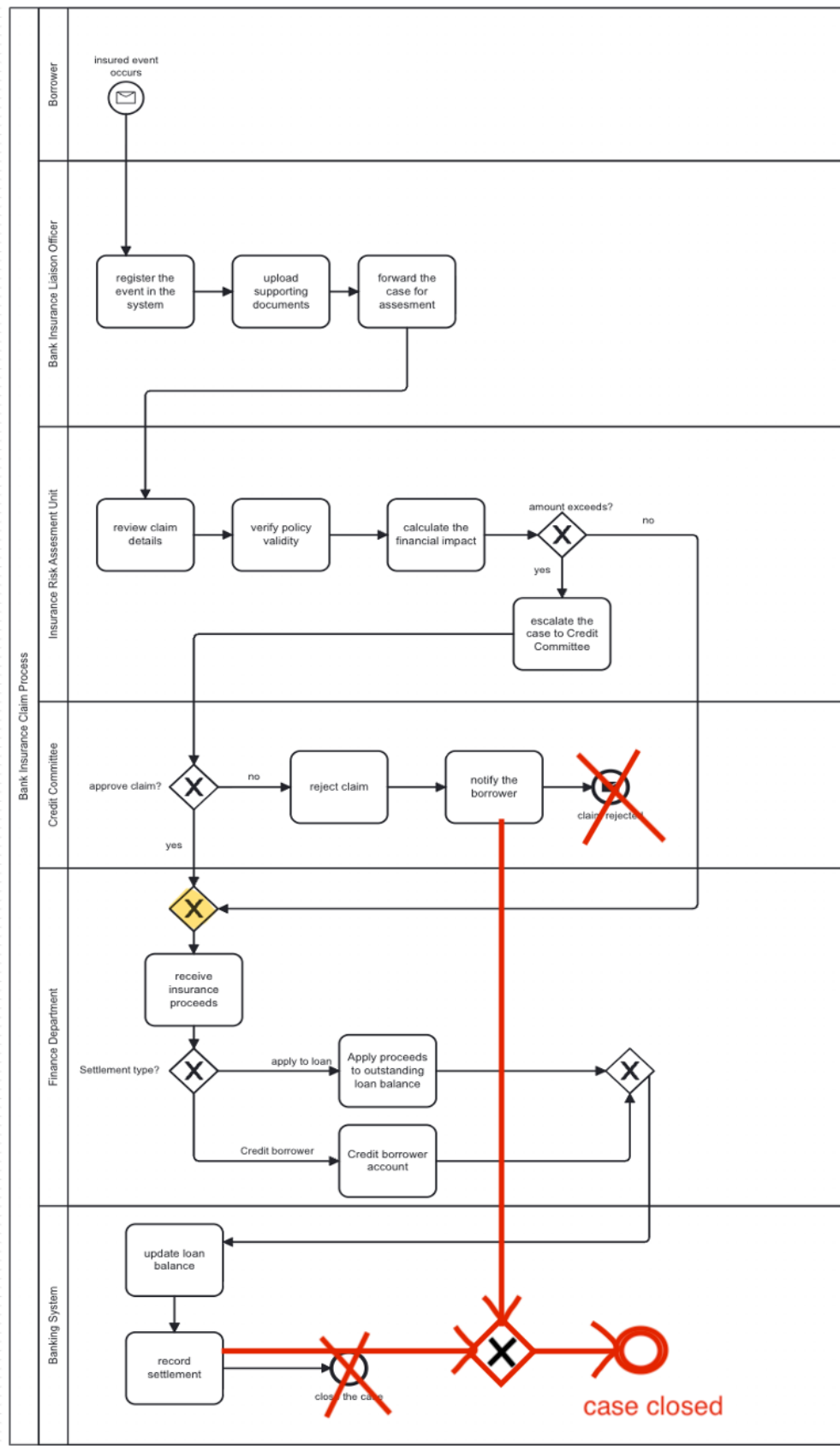
Activity	Borrower	Liaison Officer	Risk Assessment Unit	Credit Committee	Finance Department	Banking System
Register the event in the system	I	R				A
Upload supporting documents	I	R				
Forward the case for assessment		R	A			
Review claim details			R			
Verify policy validity			R			
Calculate financial impact			R			
Escalate the case to Credit Committee			R	A		
Approve or reject claim				R		
Notify the borrower	I			R		A
Receive insurance proceeds					R	
Apply proceeds to outstanding loan balance					R	A
Credit borrower account					R	A
Update loan balance						R
Record settlement						R

In case of “Register the event in the system,” the Banking System is accountable since despite the Liaison Officer being the person to enter the data, the system is the record-keeper that is authoritative and validates the registration. The Borrower is identified Informed because they are the cause of this activity.

In the case of “Notify the borrower,” the Credit Committee is checked as Responsible since the activity falls under the Credit Committee swimlane in the BPMN model, the committee triggers the notification of rejection. The Banking System is characterized as Accountable due to the fact that the notification is an automatic functioning generated by the system and sent. The Borrower is denoted Informed because he is the recipient of the outcome.

In the case of “Apply proceeds to outstanding loan balance” and “Credit borrower account,” the Finance Department is identified as Responsible since it is the human decision-maker that decides on the correct course of settlement, whereas the Banking System is identified as Accountable because it does the actual transaction and records it in the banking system.

TASK 3: SOUNDNESS CHECK



TASK 3.1. STRUCTURAL SOUNDNESS

A process model will have structural soundness when there exists exactly one start event, exactly one end event, and each node within the process model connects to a pathway that links the start to the end event. Structural soundness provides a necessary prerequisite for conducting a full behavioral soundness analysis.

There is one start event "insured event occurs" in the Bank Insurance Claims Process's AS-IS model located in the Borrower lane. However, there exist two end events "claim rejected" in Credit Committee lane and "close the case" in Banking System lane. Therefore, since there exist two end events, the model does not meet the definition of structurally sound under the formal definition.

Since we aim to transform the model without changing its behavior, the model may be changed by eliminating the two distinct end events. Next, introduce an XOR join gateway which would merge the two terminal branches. Finally, connect the XOR join gateway to a single new end event. Since the two end events are mutually exclusive, no single execution trace could ever result in reaching both end events due to all of these being separated by XOR gateways along their entire length. This transformation will provide validity to our new model and is depicted as part of the annotated diagram above.

The transformed model has every node accessible through one of the two paths from the single start event to the single end event. The start event initiates the flow into the Bank Insurance Liaison Officer lane. From there, "register the event in the system", "upload supporting documents", and "forward the case for assessment" occur in sequence as they are all accessible from the start event. Once the flow reaches the Insurance Risk Assessment Unit, "review claim details", "verify policy validity", and "calculate the financial impact" are all accessible and each flows into another activity. The flow is then routed by an XOR split to "escalate the case to Credit Committee" if yes (the amount exceeds?) and otherwise goes

directly to the XOR join and subsequently "receive insurance proceeds". This means that both branches of the flow are accessible and both will reach the end event. After entering the Credit Committee lane, the flow encounters an XOR split. If no (approve claim?), it will be directed to "reject claim" and "notify the borrower". Otherwise (yes), it will go directly forward. Thus, within this lane, both paths are accessible and both paths will proceed to the end event. When the flow reaches the Finance Department, "receive insurance proceeds" can be accessed through both incoming lanes of the flow via the preceding XOR join. Then, when encountering the XOR split "settlement type?", the flow may go either to "apply proceeds to outstanding loan balance", or to "credit borrower account". As such, within this lane, both paths are accessible, however, these paths do merge with each other at the subsequent XOR join. Lastly, once reaching the Banking System lane, "update loan balance" and "record settlement" can also be accessed and then ultimately result in the singular end event. There are no inaccessible nodes or dead paths within the model.

TASK 3.2. SOUNDNESS

Soundness has 3 main criteria:

- there will always be an eventual path to termination once a workflow begins
 - upon reaching termination the entire workflow has been completed, and no additional steps may occur
 - each step (or "activity") is included in at least one valid path within the workflow.
1. Both pathways available in the model do terminate. Once the claim amount is less than or equal to the internal authority limit, the token will proceed to the Finance Department and then to settlement before terminating. When the claim amount exceeds the internal authority limit, the case will go to the Credit Committee. Upon

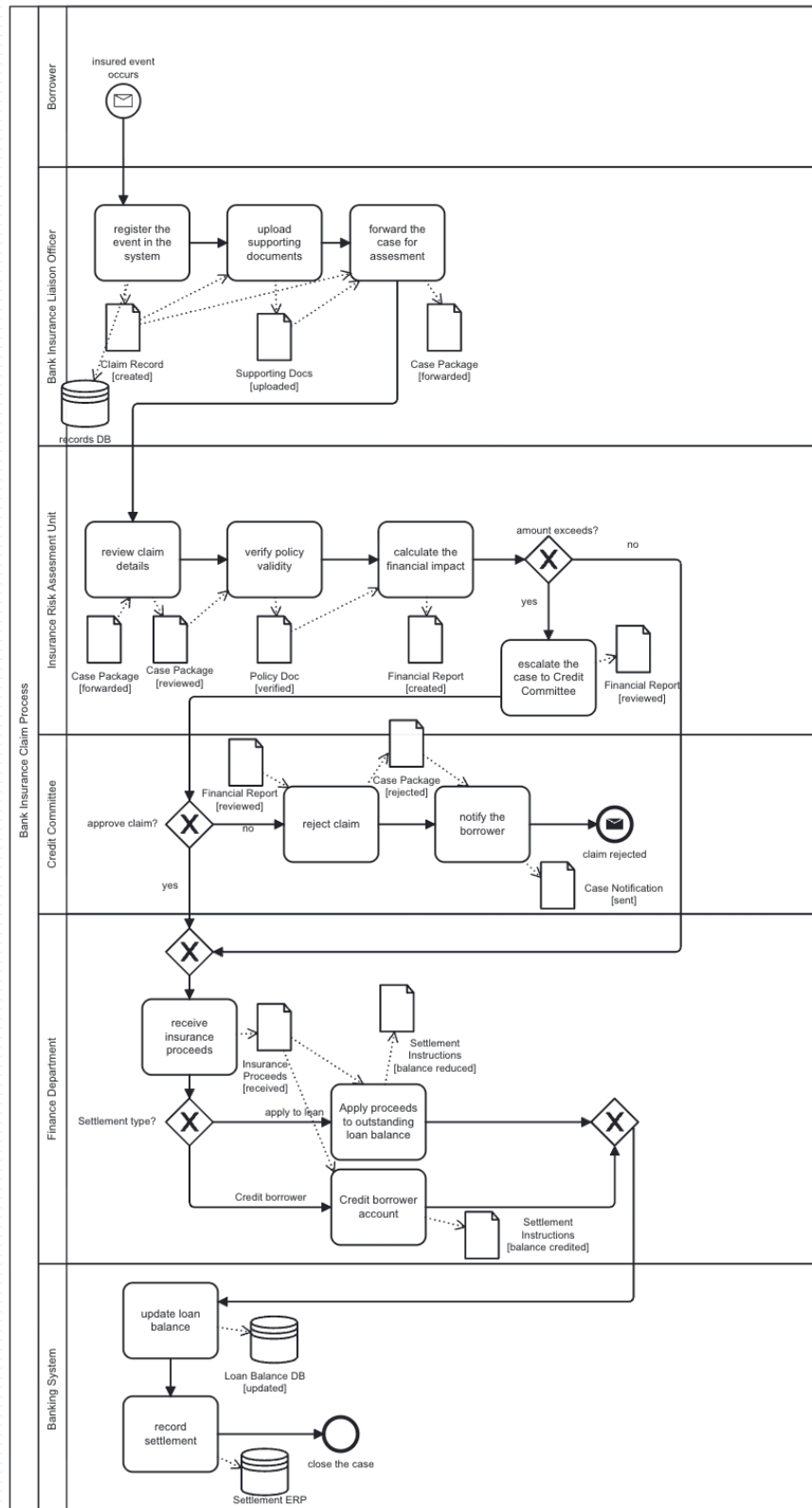
rejection by the Credit Committee, the borrower will receive notification, and the process will be terminated. When the Credit Committee approves the case, the token will proceed from the Credit Committee to the Finance Department whereupon normal termination occurs. Regardless of how the claim is processed, the process will terminate. A deadlocked state never occurs prior to the completion of all steps in the process because all XOR joins in this model are receiving precisely one input token corresponding with each preceding XOR split. There is also no AND-join in this model that would require a token to continue waiting or idle indefinitely.

2. Once "notify borrower" completes its action in the reject branch of the model there is no longer an active token in this branch. Similarly, once "record settlement" completes in the approved branch, no additional actions may occur. Because there are no loops or parallel branches within the model such that a token may remain active after the last event has occurred, there is no livelock.
3. Each activity in the model exists somewhere in a valid execution trace for the model. When the claim amount exceeds the threshold and the Credit Committee rejects the claim, "escalate case to Credit Committee", "reject claim", and "notify borrower" will be executed. Also, if the settlement type is either, "apply proceeds to outstanding loan balance" then "apply proceeds to outstanding loan balance" will be executed. Otherwise, if it is "credit borrower account" then "credit borrower account" will be executed. Finally, all remaining activities "registration", "document upload", "review", "verification", "financial impact calculation", "receiving proceeds", "updating loan balance", and "recording settlement" exist in each trace. None of these have become dead. Thus, each activity is executable.

Therefore, the model is sound. The only violation of structural soundness was the existence of two end events. As previously explained and shown with a simple example diagram which

was solved by transforming the process into one that had but one end event. No changes were needed to be made to the process logic.

TASK 4: DATA OBJECTS OF AS-IS MODEL



The diagram attached above represents the AS-IS model enriched with data objects and data stores that are generated, consumed, and updated throughout the process. Data objects were represented in Camunda modeler as “document” notations and data stores were represented using the “database” notation. Dashed arrows were used to show the inputs/outputs of data objects and data stores into/out of activities.

TASK 4.1. INPUT/OUTPUT SETS

Input/Output Sets for Each Activity:

Register the event in the system

OutputSet = {Claim Record [created]}

The activity also writes to the Records DB data store.

Upload supporting documents

InputSet = {Claim Record [created]}

OutputSet = {Supporting Docs [uploaded]}

Forward the case for assesment

InputSet = {Claim Record [created], Supporting Docs [uploaded]}

OutputSet = {Case Package [forwarded]}

Review claim details

InputSet = {Case Package [forwarded]}

OutputSet = {Case Package [reviewed]}

Verify policy validity

InputSet = {Case Package [reviewed]}

OutPut = {Policy Doc [verified]}

Calculate the financial impact

InputSet = {Policy Doc [verified]}

OutputSet = {Financial Report [created]}

Escalate the case to Credit Committee

InputSet = {Financial Report [created]}

OutputSet = {Financial Report [reviewed]}

Reject claim

InputSet = {Case Package [reviewed], Financial Report [reviewed]}

OutputSet = {Case Package [rejected]}

Notify the borrower

InputSet = {Case Package [rejected]}

OutputSet = {Case Notification [sent]}

Receive insurance proceeds

InputSet = {Approved claim} (implicit from process flow)

OutputSet = {Insurance Proceeds [received]}

Apply proceeds to outstanding loan balance

InputSet = {Insurance Proceeds [received]}

OutputSet = {Settlement Instructions [balance reduced]}

Credit borrower account

InputSet = {Insurance Proceeds [received]}

OutputSet = {Settlement Instructions [balance credited]}

Update loan balance

InputSet = {Settlement Instructions [balance reduced OR balance credited]}

OutputSet = {Loan Balance DB [updated]}

Record settlement

OutputSet = {Settlement ERP [recorded]}

TASK 4.2. OBJECT LIFECYCLE CONFORMANCE CHECK

The lifecycle check verifies that the order of states and completeness of states that an object passes through are logical and valid. Essentially, the states that a data object passes through should be in a specific order and the object should enter a terminal state (Created or Destroyed) after the last action associated with the object has occurred within the relevant process path.

Claim Record [created]

The Claim Record enters the [created] state at the start of this process due to "Register the Event in the System", and is immediately used as input to both "Upload Supporting Documents" and "Forward the Case for Assessment". The Claim Record is neither read nor written, it simply moves from one location to another. Since it is created prior to being consumed, and does not have contradictory changes in its states, the Claim Record meets all requirements to be considered to have a logical and complete lifecycle.

Supporting Docs [uploaded]

Supporting Docs are generated in the [uploaded] state as a result of "Upload Supporting Documents", and are consumed as input to "Forward the Case for Assessment". Supporting Docs have one state, which is uploaded, and they are consumed only once. They are created before they are consumed, and there are no additional state transitions on Supporting Docs after they are consumed. As such, Supporting Docs have a logical and complete lifecycle. There are no violations.

Case Package [forwarded] → [reviewed] → [rejected]

When the Case Package was initially submitted for evaluation, it presented itself to the system as the output of "Forward the case for evaluation" in the [forwarded] state. Following processing and production of an input/output by "Review Claim Details", the Case Package transitioned to its second state [reviewed]. When the Case Package proceeded

through the rejection pathway, it entered its final (third) state [rejected], as the output of "Reject Claim". The Case Package's progression from forwarded, to reviewed, to formally rejected is thus a logically sequential and consistent progression. Additionally, each of the states described above were created before they were consumed by the next action. There is no need for the Case Package to be processed again once it has proceeded along the approval pathway because the financial settlement processes that follow the approval pathway utilize no object similar to this one.

Policy Doc [verified]

The Policy Doc is created in the [Verified] state by "Verify Policy Validity" and is immediately consumed by "Calculate Financial Impact". Thus, the Policy Doc has a single state and is consumed once. Therefore, since the Policy Doc is created before it is consumed, and does not experience any conflicting transitions, the lifecycle of the Policy Doc is complete and consistent with itself. There are no violations present.

Financial Report [created] → [reviewed]

The Financial Report is originally produced in a [created] state with "calculate the financial impact". After that the report will go through the [reviewed] state as a result of having been processed by the Credit Committee after "escalate the case to Credit Committee" has been performed. There is a logical sequence of states since the report is created before it is evaluated by the credit committee. So each state is generated before it can be used for the subsequent process. The lifecycle finishes with the [reviewed] state reached after the Credit Committee have reached a final decision on approving or rejecting the case.

Case Notification [sent]

The Case Notification is initially produced in a [sent] state by "notify the borrower", which sends the notification to the Borrower indicating they have been rejected. There is one terminating state as it was both created and transmitted during this same activity. Because

there is no additional consumption or transition of states needed with regard to notifying the borrower, the lifecycle of the Case Notification object is consistent and complete. Thus, there are no violations.

Insurance Proceeds [received]

The Insurance Proceeds data object is created at the [received] state when you receive insurance proceeds and then used as an input for "Apply Proceeds to Outstanding Loan Balance" and "Credit Borrower Account". It will be produced prior to being consumed and there will be no conflicting states because the XOR gateway guarantees that only one of the two consuming activities will occur. Therefore, its lifecycle is consistent and complete. There are no violations.

Settlement Instructions [balance reduced] OR [balance credited]

The Settlement Instructions data object will be in one of two mutually exclusive states which are [Balance Reduced] from "Apply Proceeds to Outstanding Loan Balance", or [Balance Credited] from "Credit Borrower Account". Each branch represents valid final results and both produce an input for "Update Loan Balance." Because both branches are governed by an XOR Gateway, there can only be one of these states produced for a single process instance, therefore, there can never be any conflict between the two states. All transitions occur logically because the production of the state precedes its consumption, which makes them all consistent with the outcome that the decision regarding settlement were previously established.

Records DB [written]

The Records Data Base (DB) is an active database that records "the event in the system" and writes the Claims Record at the beginning of the process and retains it beyond the life of this process instance. Because it is a database, it does not get used or sent between processes, so it remains independent. Only one time is there a write interaction for the

Records DB when it was first opened. This activity is consistent with the purpose of the Records DB as a long-term audit repository. There are no violations.

Loan Balance DB [updated]

The Loan Balance Data Base is also a long-term database that is written to "update loan balance," changing from the [not updated] to the [updated] state once the settlement amount has been added to the loan account. After consuming the settlement instructions object, the Loan Balance database has a write occurrence. Again, because this is a data base and not a data object, it will persist for longer than the duration of this process and will be available for use by future processes but in a different way. Following the decision on how much of the settlement should be deducted from the borrowers' loan accounts, there is a single write interaction that fits perfectly with its function as the official record of the borrower's loan accounts. There are no violations.

Settlement ERP [recorded]

The Settlement Enterprise Resource Planning (ERP) is a long-term database which records "settlement" and changes from the [not recorded] to the [recorded] state as part of finalizing the process. The writing of the Settlement ERP occurs at the proper last step of the process after all financial changes have been made and recorded into the Loan Balance DB. Also, like the Loan Balance DB, it is a database and therefore can remain active for a longer period than the life of this process. A single write interaction will occur at the completion of this process. Like other aspects of this process, this write activity is consistent with its role as the official recording of the settlement being finalized. There are no violations.

CONCLUSION

The report has produced an ordered, formally based analysis of the Insurance Event Application process. It included refining the AS-IS model from stage 1, mapping of all

organizational responsibilities to activities in the model, verification that the model was correct both structurally and behaviourally, and documentation of all aspects of the data landscape associated with this process.

The Process Analysis indicated that the process is organized, roles are clear, there will be neither deadlock nor livelock potential present during process execution, and there will be consistency throughout the process with respect to the flow of data. Formal issues identified as a result of analysis included only one that there are two end events. A single structural transformation was applied to the model as part of the analysis in order to resolve this issue, confirming that the model satisfies all relevant soundness properties. Each activity in the process follows role-based allocation, all data objects within the process transitioned through logically valid states without violating their respective lifecycle.

In general, the Insurance Event Application Process is organizationally grounded, behaviourally correct, and data-consistent.

TASK BREAKDOWN TABLE

Team Member	Contribution to project	Estimated %
Ulkar Ahmadli	Soundness Check; Data Objects of As-Is Model	33.3%
Team leader: Seljan Musayeva	Stage 1 Reflection; Data Objects of As-Is Model	33.3%
Farid Aslanli	Resource Allocation Annotation; RACI & Resource Allocation Table	33.3%
Lutvu Aliyev	Didn't join groupwork	0%